TABLE I. Pump beam parameters.

Beam	w_0 (mm)	z_c (mm)
1	0.062	178
2	0.067	213
3	0.072	251
4	0.085	298
5	0.095	355
6	0.105	422
7	0.120	510
8	0.142	635

and Δy_1 for any $z > L$. Alternativelly, assuming any fixed value for $\mathbf{q}_2$, Eq. (16) allows us to calculate Δk_{1x} and Δk_{1y}, which do not depend on z. As an example, Fig. 3 shows plots of Δx_1 and Δy_1 for $\boldsymbol{\rho}_2 = 0$ and plots of Δk_{x1} and Δk_{y1} for $\mathbf{q}_2 = 0$, for a 5 mm-long BBO crystal pumped by 355 nm gaussian laser beams whose waists w_0 and waist positions z_c are shown in Table I. Uncertainties Δx_1 and Δy_1 where calculated on the plane $z = 5.0001$ mm.

One can see that while the position uncertainties in the x and y directions are about the same, the momentum uncertainties are strongly affected by the anisotropy. This is a direct consequence of the partial transfer of the pump beam angular spectrum in the direction parallel to the principal plane [19], in our case, the x direction.

It is interesting to note that the uncertainties Δx_1 and Δy_1 increase rapidly as the distance from the crystal increases in the z direction. In general, position uncertainties depend on the pump beam parameters, as exemplified in Fig. 4. For a weakly focused laser beam ($w_0 = 0.5$ mm) whose waist is located at the crystal input face ($z = 0$), our predictions shown in Fig. 4a are in good agreement with the results reported in Ref [17]. This agreement is because in this case a Gaussian model fits well the uncertainty in position coordinates [20].

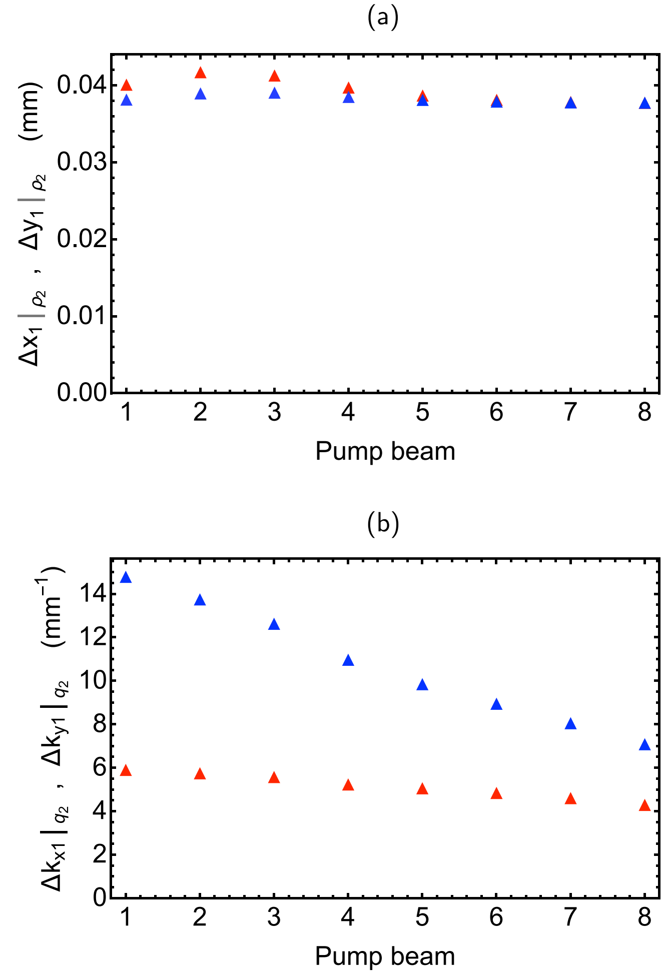

FIG. 3. Predicted values of (a) $\Delta x_1|_{\boldsymbol{\rho}_2=0}$ (red) and $\Delta y_1|_{\boldsymbol{\rho}_2=0}$ (blue), (b) $\Delta k_{x1}|_{\mathbf{q}_2=0}$ (red) and $\Delta k_{y1}|_{\mathbf{q}_2=0}$ (blue) for a 5 mm-long BBO crystal pumped by 355 nm laser beams whose parameters are listed in Table I.

IV. EXPERIMENT

EPR correlations predicted in the previous section were tested experimentally with the setup represented in Fig. 5. A 5 mm-long BBO crystal cut for type I collinear phase match (NLC) having its optic axis parallel to the xz plane and input face located on the plane $z = 0$ was pumped by a 355 nm laser beam polarized in the x direction, propagating along the z direction. The beam parameters, listed in Table I, were changed with the help of a telescope composed by a lens L_3 of focal length 50 mm and a lens L_4 of focal length 40 mm separated from L_3 by a variable distance. The down-converted light, with $\lambda_1 = 690$ nm and $\lambda_2 = 731$ nm was sent to a beam splitter (BS) and directed to detectors D_1 (equipped with a 12 nm band-pass filter centered at 690 nm) and D_2 (equipped with a 40 nm band-pass filter centered at 730 nm). Lenses L_1 and L_2 of focal length 75 mm were placed at 150 mm from the output face of the nonlinear crystal. Detectors D_1 and D_2 were placed at 75 mm from L_1 and L_2 (Fourier plane) for the measurements of $|\psi(\mathbf{q}_1, 0, L)|^2$ and at 150 mm (1:1 image plane) for the measurements of $|\psi(\boldsymbol{\rho}_1, 0, L)|^2$. Each detector consists of a multimode optical fiber with a diameter of 50 μm, one tip mounted on a computer-controlled xy motorized translation stage and the other tip coupled to a photon-counting avalanche photodiode. In all measurements, D_2 was kept at $\boldsymbol{\rho}_2 = 0$.

Due to the relatively large fiber diameter and photon-counting fluctuations, experimental results are not directly comparable with theory. To check the accuracy of theoretical predictions, the following procedure was adopted: Experimental data for coincidence detections on the image plane were fit to a hyperbolic secant distribution $A\,\mathrm{sech}\,\pi\xi/2\Delta_\xi$ ($\xi = x, y$), whose standard deviation is given by Δ_ξ. Numerical convolutions of the theoretical coincidence profiles with the 50 μm circular

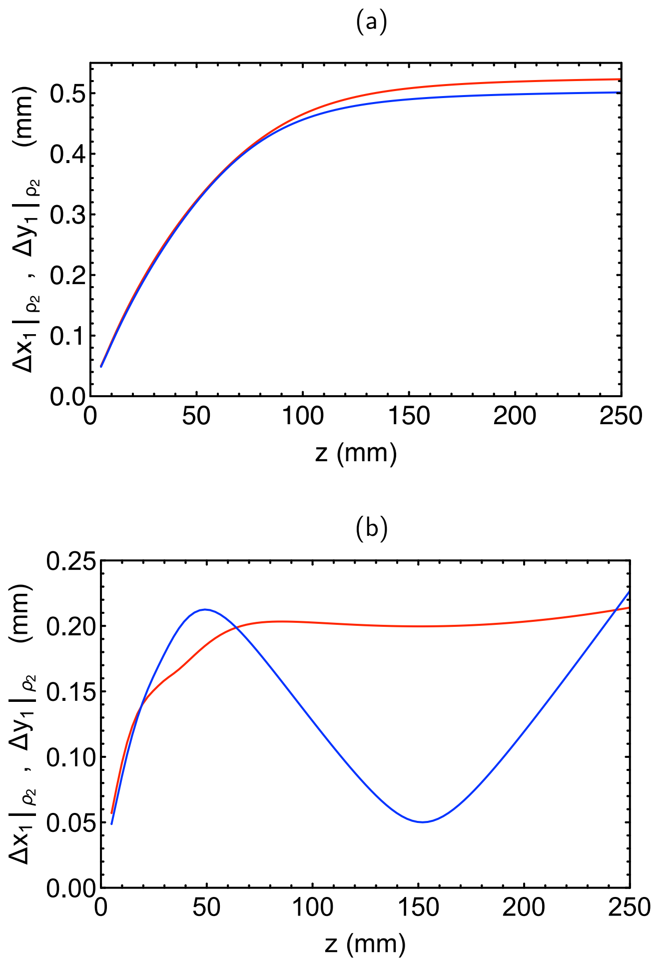

FIG. 4. Predicted values of $\Delta x_1|_{\rho_2=0}$ (red) and $\Delta y_1|_{\rho_2=0}$ (blue) as functions of the distance from the output face of a 5 mm-long BBO crystal cut for collinear degenerate phase match, pumped by a 355 nm laser beam with (a) $w_0 = 0.5$ mm, $z_c = 0$ and (b) $w_0 = 0.05$ mm, $z_c = 150$ mm.

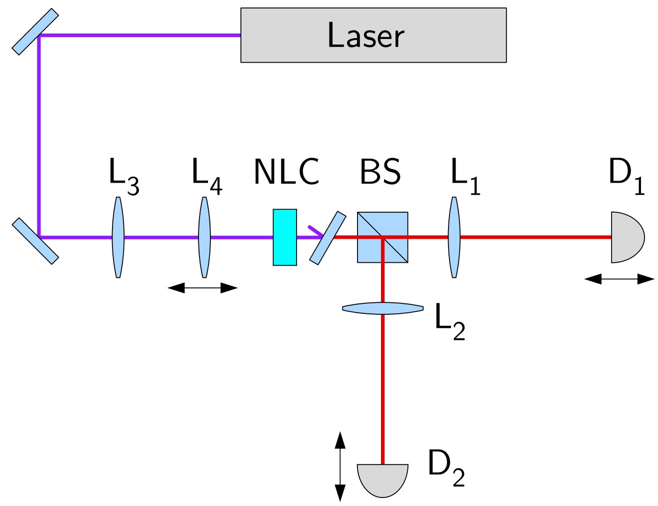

FIG. 5. Experimental setup

apertures of D_1 and D_2 were made, resulting in the expected detection probability distributions. Profiles obtained when the crystal was pumped with the laser beam #1 (see Table I) are shown in Fig. 6. An additional correction of the beam waist radius was made, due to the pump beam M^2 factor of 1.14. On the Fourier plane, a Gaussian distribution $A\exp(-\xi^2/2\Delta_\xi^2)$ $(\xi = k_x, k_y)$ was used in a similar procedure. Experimental results and theoretical predictions are presented in Fig. 7.

V. DISCUSSION AND CONCLUSION

From the results presented here, one can see that the EPR correlations $\Delta x \Delta k_x$ and $\Delta y \Delta k_y$ of the photons pairs generated by spontaneous parametric down-conversion are strongly affected by the pump beam angular spectrum in the direction normal to the principal plane (defined by the optic axis and the z axis). Such dependence is much smaller in the direction parallel to the principal plane. This effect is readily explained by the presence of the term $\operatorname{sinc} l_t Q_x$ in Eq. (16). That term, which depends on the birefringence, the phase match angle, and the crystal length (see Eq. (3)), acts as a spatial filter for the transfer of the angular spectrum from the pump beam to the two-photon state [19]. Because of this filtering effect, the product $\Delta x \Delta k_x$ behaves like the laser beam was less focused. The two uncertainty products $\Delta x \Delta k_x$ and $\Delta y \Delta k_y$ tend to a unique minimum value as the pump beam gets more collimated, that is, $w_0 \gg l_t$. In our case, $l_t = 0.186$ mm.

In conclusion, we have presented, unlike any previous publication: (a) Accurate analytic expressions for the coincidence detection probability amplitudes of photon pairs generated by spontaneous parametric down-conversion in both momentum and position spaces on the entire plane normal to the pump beam. Those expressions allow us to predict how the correlations in position and momentum depend on the system parameters like crystal length, crystal birefringence, pump beam focusing, pump beam waist location, and detectors locations. (b) Experimental data supporting our theoretical predictions, using Einstein-Podolsky-Rosen correlations as benchmarks, for 8 different pump beam configurations.

The results presented here may be useful in any application relying on position and momentum correlations of photon pairs generated by SPDC in birefringent crystals.

ACKNOWLEDGMENTS

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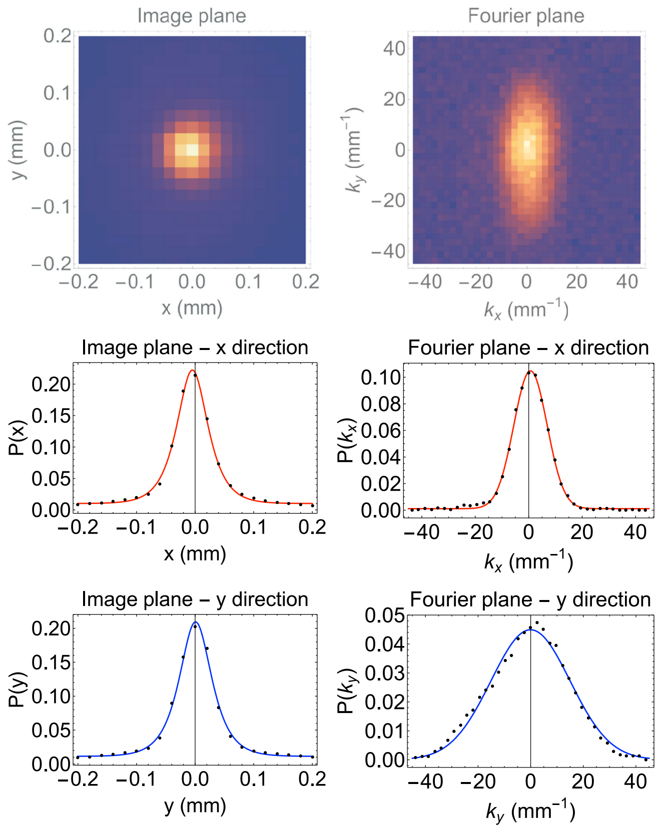

FIG. 6. Top row: Examples of coincidence detection profiles on the image plane (left) and on the Fourier plane (right) for beam # 1 (see Table I). Mid row: Corresponding detection probability densities $P(x_1|\boldsymbol{\rho}_2 = 0)$ (left) and $P(y_1|\boldsymbol{\rho}_2 = 0)$ (right). Bottom row: Corresponding detection probability densities $P(k_{x1}|\mathbf{q}_2 = 0)$ (left) and $P(k_{y1}|\mathbf{q}_2 = 0)$ (right). Dots are normalized experimental data and solid lines are best fits for distributions $A\,\mathrm{sech}\,\pi\xi/2\Delta_\xi$ ($\xi = x, y$ on the image plane) and $A\exp(-\xi^2/2\Delta_\xi^2)$ ($\xi = k_x, k_y$ on the Fourier plane).

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